

Matthew Topham

AI Systems & Full-Stack Developer

Boulder, Colorado · thetopham@gmail.com
thetopham.github.io · github.com/thetopham · linkedin.com/in/mrtopham

Computer science student at CU Boulder building AI applications, wearable assistants, and local AI infrastructure. Open-source work spans full-stack development, persistent state, model integrations, and regression testing.

OPEN-SOURCE EXPERIENCE

Marinara Engine + Agents · Pasta Devs

Open-Source Developer · 2026-Present

- Contributed 100 merged pull requests across Engine (68) and Agents (32), spanning React/TypeScript UI, Node.js services, capability APIs, persistence, and automated tests.
- Developed hierarchical world state, validated movement, snapshots/checkpoints, and versioned host interfaces for model routing, resources, and transactional persistence.
- Connected roleplay exchanges, visual references, keyframes, image-aware motion planning, and ComfyUI image-to-video generation; added regression and browser-test coverage.

Selected work: [animation pipeline](#) · [world state](#) · [capability API](#)

SELECTED PROJECTS

Manfred · Wearable AI assistant

Creator · 2026 · Development prototype; smart-glasses image capture is experimental

- Developed an Android/Flutter companion and Python services for BLE audio ingestion, durable upload buffering, transcription, searchable archives, and experimental image capture.
- Separated wearable services from knowledge and compute components using explicit export contracts and independent source/Android validation.

Demerzel · Local AI task orchestration

Creator · 2026 · Development project; worker dispatch paused pending post-refactor acceptance

- Developed durable task queues, bounded worker execution, lifecycle monitoring, and reviewable artifacts for local AI workloads.
- Separated knowledge, task execution, and fleet control behind validated interfaces; implemented review bundles and an operator-facing control center.

School of the Ancients · AI / VR learning

Built voice-driven Socratic learning experiences with React/TypeScript, Node.js, Supabase/Postgres, real-time voice integration, and Unity XR prototypes.

TECHNICAL SKILLS

Languages & applications: TypeScript, JavaScript, Python, SQL, React, Node.js, Flutter, REST, WebSockets.

Systems & validation: Linux, Docker, GitHub Actions, Playwright, local inference, ComfyUI, transactions, retries, regression testing.

EDUCATION & CERTIFICATION

University of Colorado Boulder - B.A. Computer Science, expected May 2027

Front Range Community College - A.S. Computer Science, 2025

NVIDIA-Certified Associate - AI Infrastructure & Operations, August 2025